

Independent Cohort Gaze Comparison

A concise framework for comparing 500 Webcam/Prolific participants with 500 different professional-camera participants.

500 Webcam	Different people	500 Professional
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What this design can answer

It estimates whether the complete Webcam workflow-and-cohort distribution is sufficiently similar to the professional workflow-and-cohort distribution for a named group-level endpoint. Equal sample size improves precision, but does not remove differences in recruitment, setting, display, lighting, or participant composition.

It cannot estimate individual device agreement, device interchangeability, or a pure causal device effect because no participant is measured by both workflows.

Four-step analysis

1	Comparability Verify identical stimulus, task, exposure, coordinate transform, and QC versions. Review participant balance with SMD.	2	Technical quality Compare participant-level accuracy, precision, data loss, and valid-trial share using Welch intervals and equivalence margins.
3	Group attention Compare cross-domain heatmaps against each cohort's own repeated split-half stability using histogram intersection.	4	Domain detectability Use repeated cross-validated logistic regression to test technical and attention feature spaces separately.

1. Comparability

For a prespecified participant characteristic X , measure balance with the standardized mean difference:

$$\text{SMD} = (\text{mean}_W - \text{mean}_P) / s_{\text{pooled}}$$

A practical review flag is $|\text{SMD}| \geq 0.10$. This is not proof of confounding or randomization. Acquisition-context variables such as display size and lighting describe the real workflow and should not be silently adjusted away.

2. Technical quality and equivalence

Compute one value per participant. Calibration accuracy is mean distance to known targets. Precision is RMS consecutive displacement at a stable target. Data loss is invalid samples divided by expected samples. Valid-trial share is valid trials divided by expected trials.

$$\begin{aligned} \Delta &= \text{mean}_W - \text{mean}_P \\ \text{SE} &= \sqrt{s_W^2/n_W + s_P^2/n_P} \\ \text{CI}_{90} &= \Delta \pm t(0.95, \text{Welch df}) \times \text{SE} \end{aligned}$$

With a prespecified acceptable difference M , support equivalence only when the entire 90% interval is inside $(-M, +M)$. A non-significant difference is not evidence of equivalence.

Group attention and domain diagnostics

The same-domain question is separated into technical acquisition and attention-pattern endpoints.

3. Group attention maps

For participant i and map bin b , sum valid dwell duration d over fixations that fall in that bin. Aggregate participants, apply one common smoothing rule, and normalize the map to sum to 1.

```
H_i(b) = sum_j d_ij x I(g_ij is in bin b)
P_G(b) = Smooth(sum_i H_i(b)) / sum_b Smooth(sum_i H_i(b))
```

Map similarity uses histogram intersection. It is the shared normalized density between two maps:

```
SIM(P,Q) = sum_b min(P(b), Q(b)), with 0 <= SIM <= 1
```

Repeatedly split each cohort into two random halves. Compute Webcam split-half SIM, professional split-half SIM, and cross-domain SIM. The comparison is:

```
gap = SIM_cross - min(SIM_W_split, SIM_P_split)
Support noninferiority when the lower 90% split-range is above -M_map.
```

This uses each workflow's ordinary participant-sampling variability as a realistic benchmark instead of demanding a perfect SIM of 1. For real inference, add a participant bootstrap with replacement for the population cross-domain interval.

4. Domain detectability

Set $Y=1$ for Webcam and $Y=0$ for professional. Fit two separate regularized logistic models: one using technical-quality features and one using stimulus-by-AOI dwell shares. Standardize with training-fold statistics only.

```
Pr(Y=1|X) = 1 / [1 + exp(-(beta_0 + sum_j beta_j z_j))]
AUC = Pr(score_W > score_P) + 0.5 x Pr(tie)
```

Use repeated stratified five-fold cross-validation and report an out-of-fold AUC with a participant-bootstrap interval. AUC 0.50 means chance discrimination; higher values mean greater separability. Low AUC supports low detectability, but never proves equality.

How the decisions fit together

Step	Question	Decision output
1	Comparable protocol and cohorts?	Pass or stop/restrict
2	Technical differences within margins?	Similar / different / inconclusive
3	Cross-domain map loss acceptable?	Noninferior to split-half baseline
4	Can source domain be predicted?	Technical AUC and attention AUC

Synthetic example only

Technical AUC 0.995: workflows are technically distinguishable.
Attention AUC 0.552: attention features have low source detectability.
Cross-domain SIM 0.954-0.961 vs within-cohort SIM about 0.963-0.964: group maps are close to split-half repeatability under the illustrative margin.

Defensible conclusion: For identical stimuli and a harmonized protocol, Webcam group maps may be sufficiently close to professional-workflow maps for a prespecified group-level endpoint. This does not establish individual interchangeability or a pure device effect.

Methods basis: Lakens (equivalence); Lopez-Paz and Oquab (classifier two-sample tests); Kummerer et al. (saliency metrics). Synthetic methods only; not a Mayo clinical validation result.